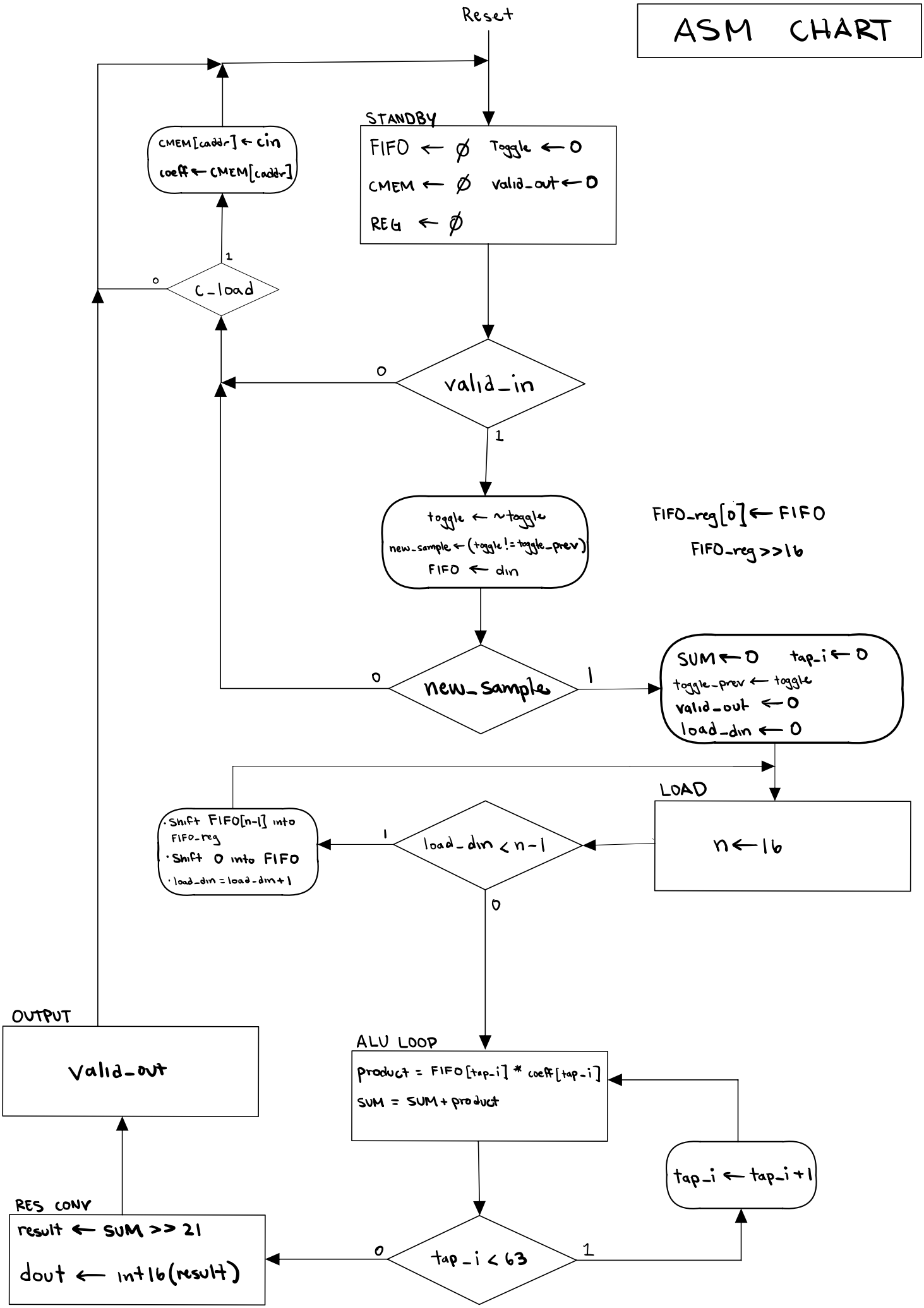
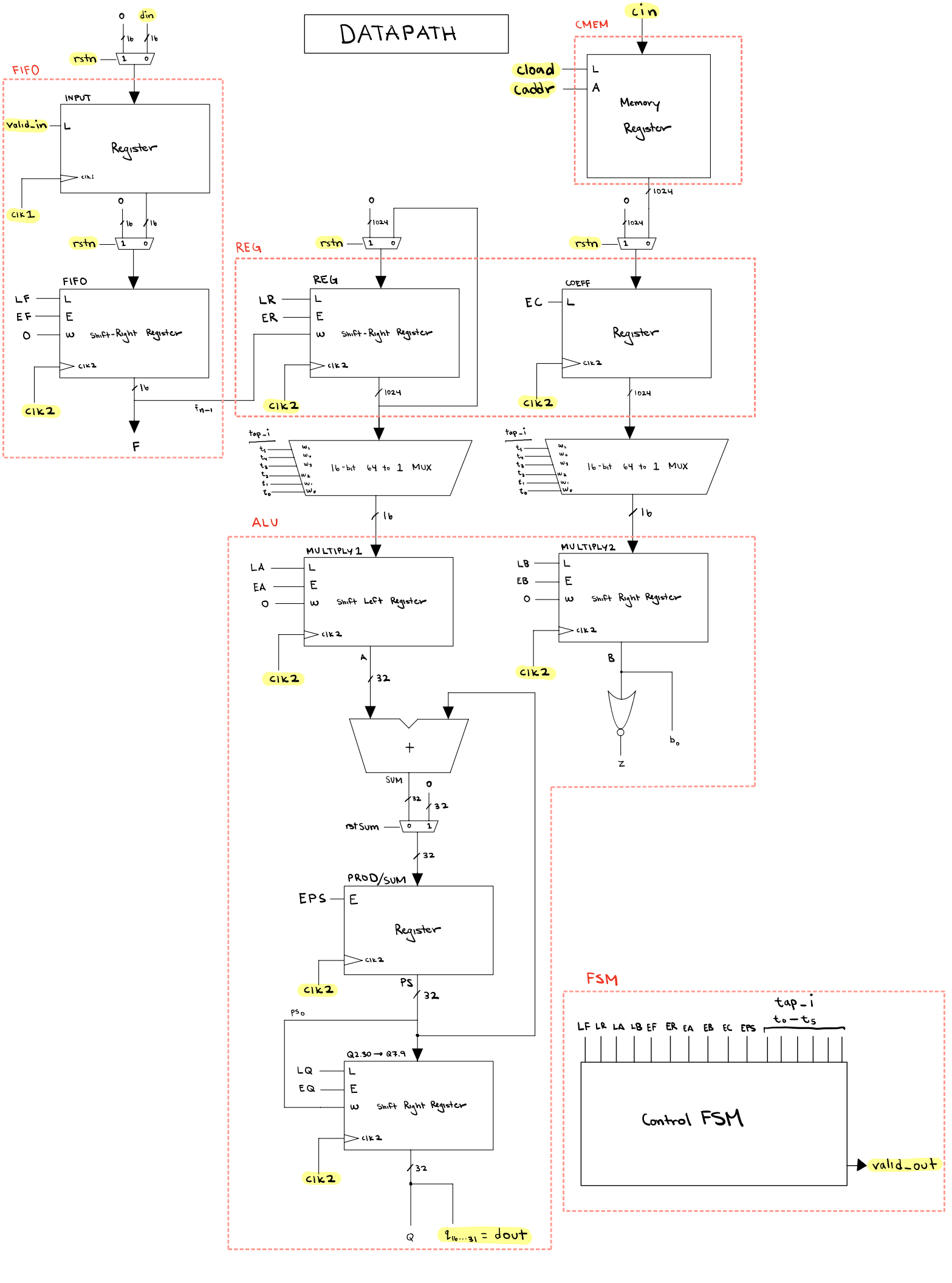
ASM CHART


Reset
STANDBY
FIFO ← ∅ Toggle ← 0
CMEM ← ∅ valid_out ← 0
REG ← ∅
c_load
1
0
CMEM[caddr] ← cin
coeff ← CMEM[caddr]
valid_in
0
1
toggle ← ~toggle
new_sample ← (toggle != toggle_prev)
FIFO ← din
FIFO_reg[0] ← FIFO
FIFO_reg >>16
new_sample
0
1
SUM ← 0 tap_i ← 0
toggle_prev ← toggle
valid_out ← 0
load_din ← 0
LOAD
n ← 16
load_din < n-1
1
0
· Shift FIFO[n-1] into FIFO_reg
· Shift 0 into FIFO
· load_din = load_din+1
ALU LOOP
product = FIFO[tap_i] * coeff[tap_i]
SUM = SUM + product
tap_i ← tap_i +1
tap_i < 63
0
1
RES CONV
result ← SUM >> 21
dout ← int16(result)
OUTPUT
valid_out


DATAPATH
FIFO
INPUT
Register
valid_in
L
clk1
rstn
0
din
16
1
FIFO
LF
EF
0
E
w
Shift-Right Register
clk2
F
fn-1
CMEM
cin
cload
caddr
A
Memory Register
1024
REG
LR
ER
Shift-Right Register
COEFF
EC
tap_i
16-bit 64 to 1 MUX
ALU
MULTIPLY1
LA
EA
Shift Left Register
A
32
MULTIPLY2
LB
EB
Shift Right Register
B
z
b0
+
SUM
rstSum
PROD/SUM
EPS
PS
PS0
Q2.30 → Q7.9
LQ
EQ
Q
f16...31 = dout
FSM
LF LR LA LB EF ER EA EB EC EPS
t0 – t5
Control FSM
valid_out